

ReachNT

Urgent repairs fixed on time out bush, without letting cost quietly decide who waits

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Remote repairs cost more, so they quietly fall to the back

70

remote communities in our model, with 4,481 public houses

8.5×

what an emergency repair costs in a very remote community, against a remote one (Nous Group, 2017)

96%

of an emergency job's cost can be travel alone (Nous Group, 2017)

6 h

drive from Katherine to Kalkarindji, one of the 16 communities its tradespeople serve



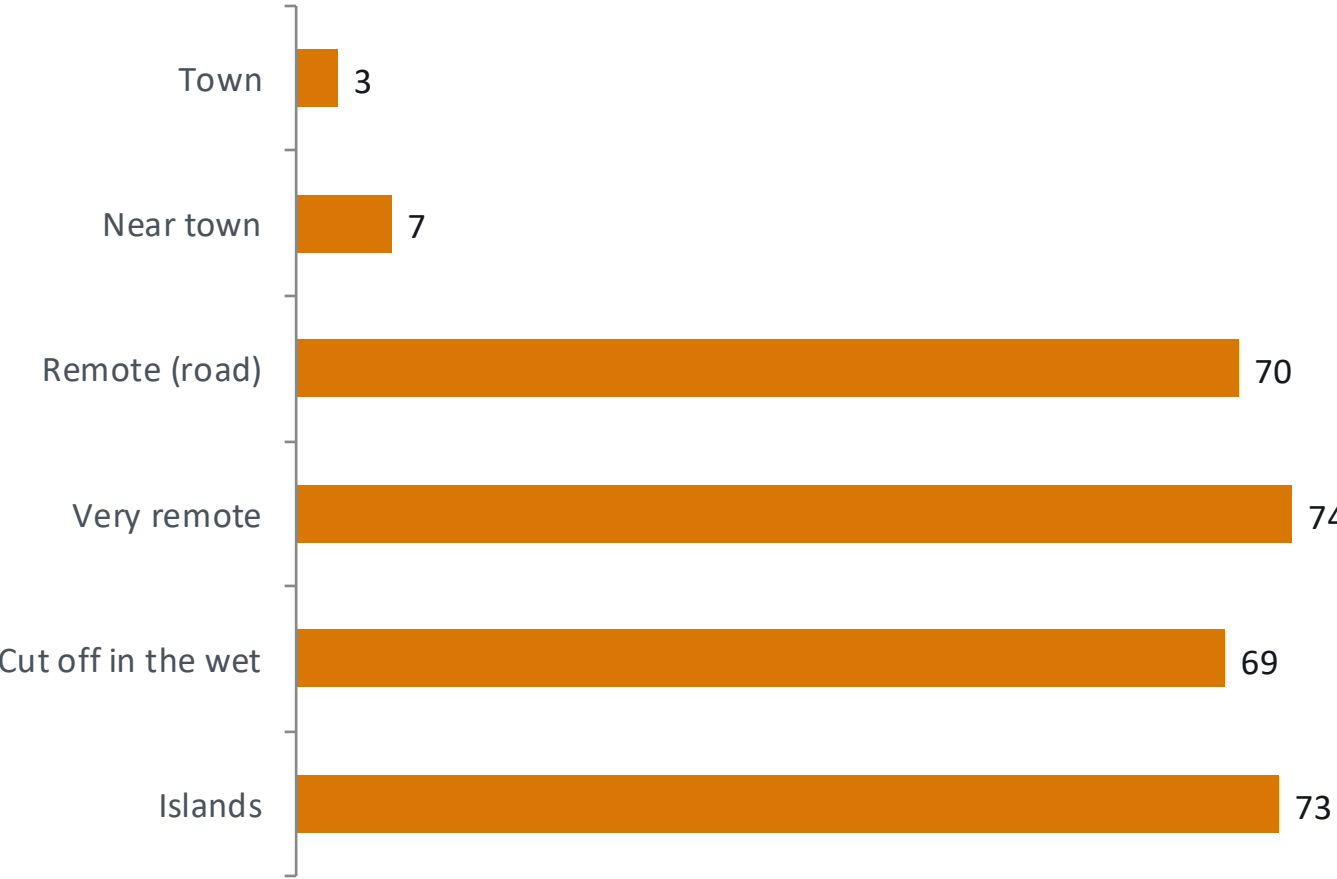
A real kind of message, from our simulated year

“Your no water to the house repair is urgent. Waiting 57 days. Held up by travel cost.”

For 28 days no plumber was sent to your community. It costs about \$7,700 to fly one there and back. That was a cost decision, not a judgement about your repair. Nobody signed off on it.

Plan for the cheapest jobs, and remote tenants wait weeks

Cheapest jobs first: days until 9 in 10 urgent repairs are fixed



62 days

for 9 in 10 urgent remote repairs, against 3 days in town

Why: a schedule built to fix the most jobs for the money always picks town first. It's efficient on paper, and no one ever decides that remote families should wait.

ReachNT: rank by need, share trips, and put a name on the trade-off



Read

Reads what the tenant said, in their words, and picks the fault and how urgent it is. Unsure? A person calls back.



Rank by need

Danger, health, who lives there and time waited set the order. Distance and cost are never in the score.



Plan shared trips

Each week it plans who goes where. Neighbouring communities share one trip, found with Uber's H3 hexagons.



Explain and sign

A named coordinator signs how much cost may count. Every tenant can see why their repair waited.

One tool, three views: the coordinator plans, the tradesperson gets a run sheet that works without signal, and the tenant gets a straight answer.

AI that reads tenants' words, and knows when to ask a person

What the tenant said

"no sparks now but the switch is black and kids touch it"

Word rules + a learning model

Rules catch known phrases; a logistic regression model catches wording the rules miss

Sure, and not dangerous?

Ranked straight away

Unsure or risky?

A person checks today. It never downgrades danger on its own

99.3%

of dangerous reports caught, on wording the AI had never seen, because doubtful ones go to a person

0.84

ROC-AUC for the model alone on new wording (1.00 on familiar)

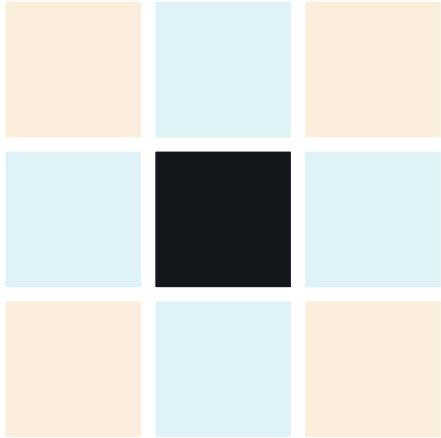
40%

of unfamiliar reports sent to a person

We never show tenants a confidence percentage: when the model said it was 90% sure, it was right about half the time. Being honest about that is part of the design.

Why we built it on Uber's H3 hexagon grid

Squares



8 neighbours at 2 different distances

Hexagons



6 neighbours, all the same distance away

Why it matters: “two hexagons away” is the same distance in every direction, so deciding which communities can share a trip is fair to all of them.



Shared trips

Communities within 2 hexagons (about 90 km) can share a trip, saving \$66–\$73 per repair



Privacy

A house is stored as a small hexagon, not a street address



Public numbers

Waits are published per larger hexagon, only where there are at least 5 repairs

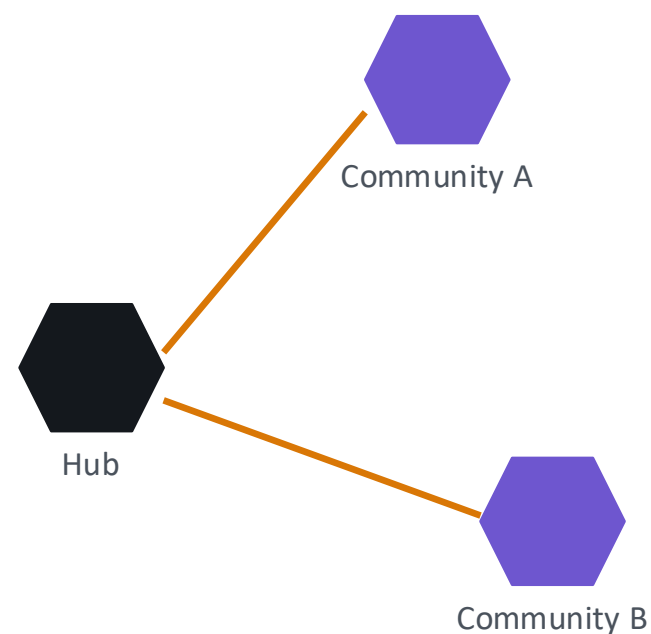


Field and maps

"Nearby jobs" for tradespeople, the driving-time map, and fast database lookups

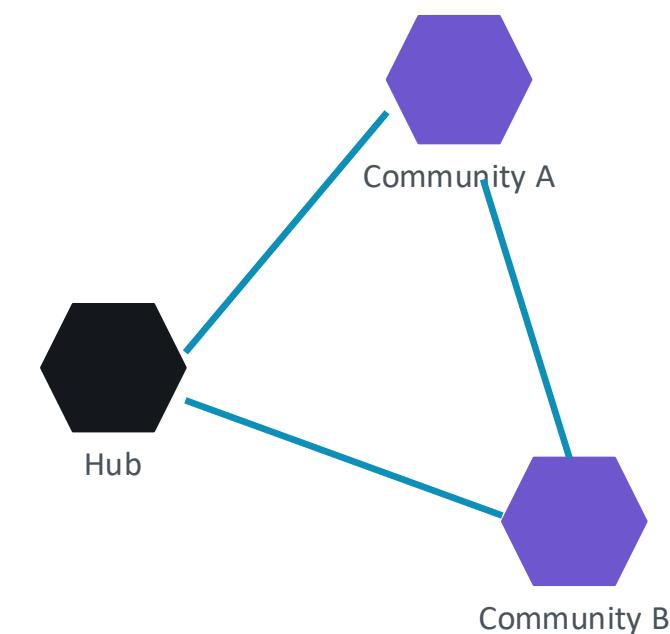
Every week, an optimiser plans the trips

Two separate trips



Two drives there and back

One shared trip



One loop: hub → A → B → hub

\$73

saved per repair by shared trips, against the same plan without them

18%

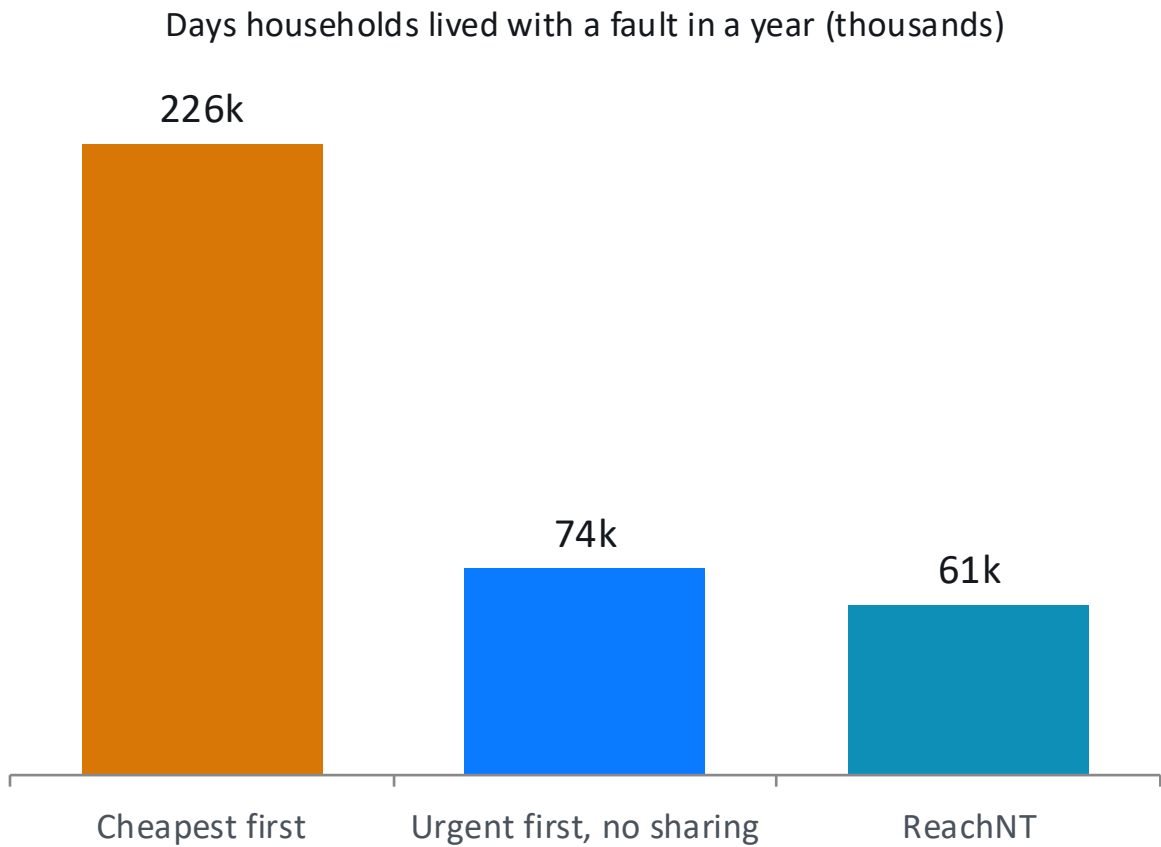
fewer days households live with a fault, because the saving buys more visits

99.4%

of 25,996 weekly plans proved the best possible, in milliseconds

Google OR-Tools (CP-SAT) chooses which communities get a visit and which jobs are done, within each tradesperson's 40 hours, road closures and airstrips.

Fairness costs a little more, and saves months of waiting



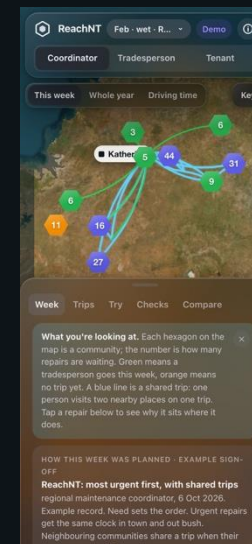
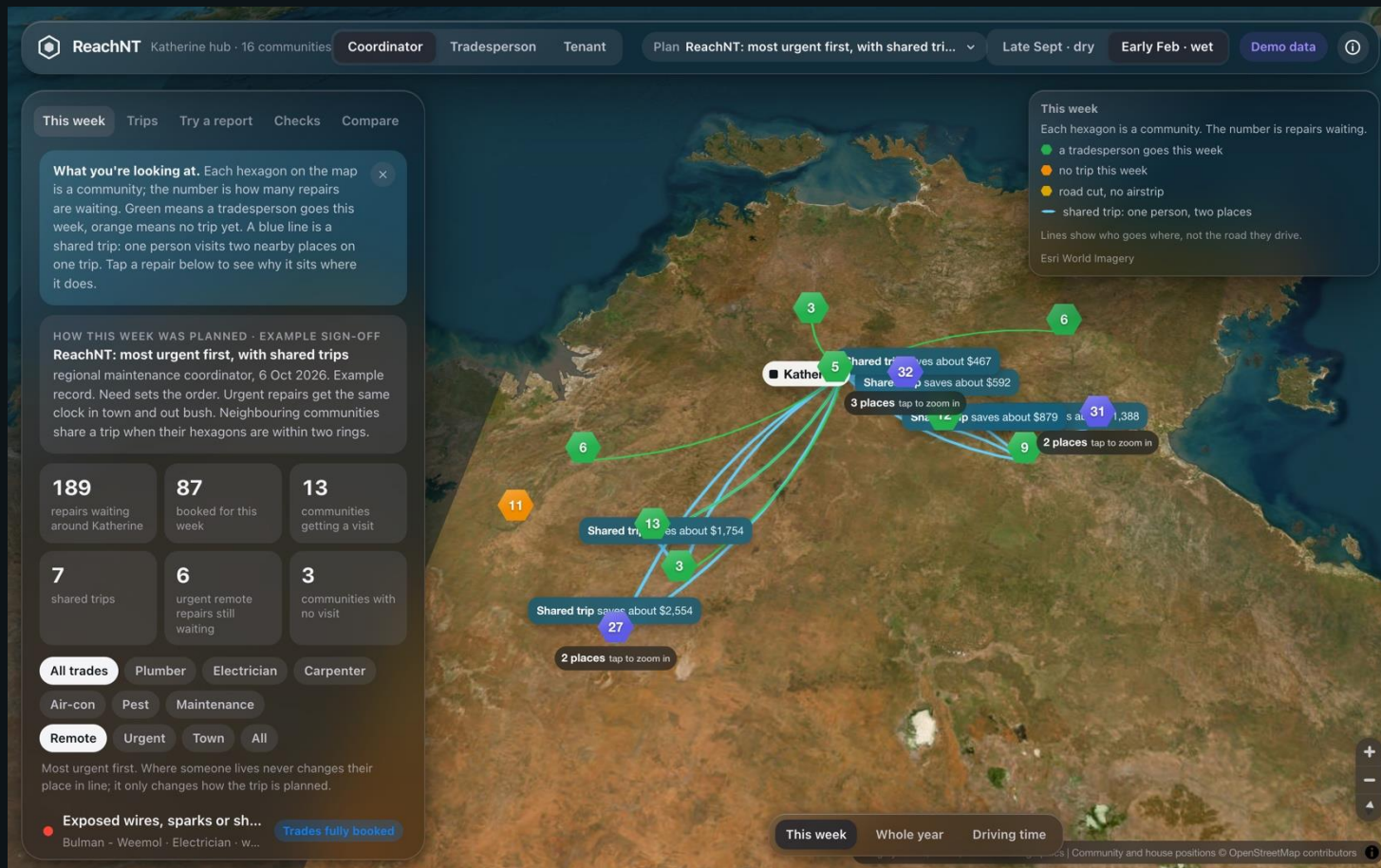
	Cheapest first	ReachNT
Average cost per repair	\$559	\$803
9 in 10 urgent remote repairs fixed within	62 days	3 days
Days living with a fault	226k	61k

+44% cost, 73% fewer days with a fault

It held in all five random years we tested: 73%–75% fewer fault-days, for 42%–46% more per repair.

One simulated year on real NT geography: 37,508 synthetic repair requests, 70 communities, 5 trade hubs.

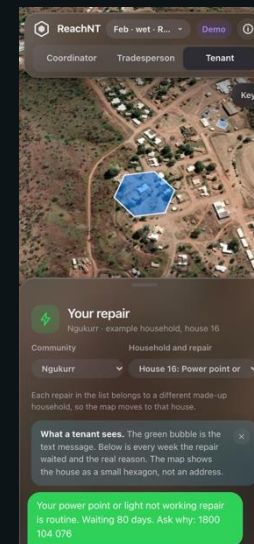
Live now, on a real satellite map, and on any phone



Coordinator



Tradesperson

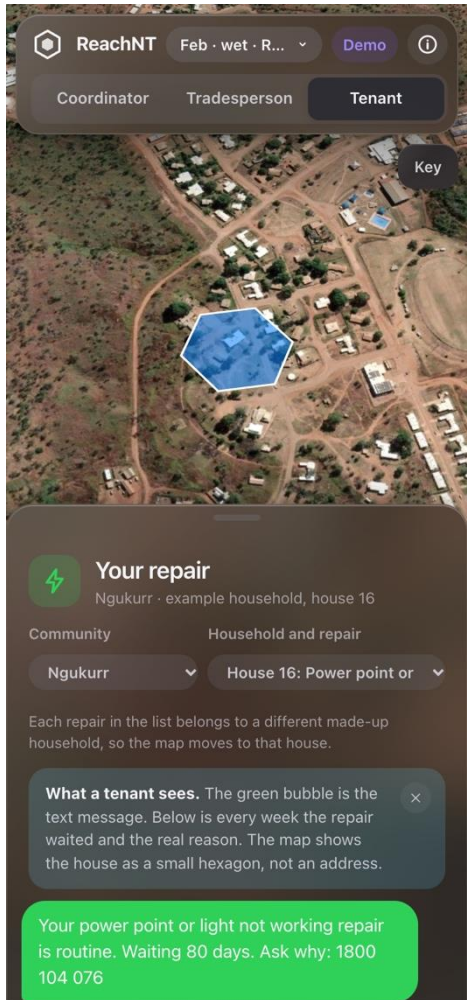


Tenant

- Works without signal: the run sheet saves updates on the phone and sends them later
- Every repair and run sheet saves as a PDF
- Plain words, WCAG 2.2 AA checked, no confidence percentages

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People stay in charge, and tenants can push back



A named person signs the trade-off

How much cost may count is a signed decision, kept in a tamper-evident record



Tenants can ask for a review

A person, not the computer, answers within 10 working days



People check the AI every week

1 in 20 reports the computer read alone is re-read by a person



No job closed on "no one home"

The time and what was tried are recorded; the tenant is told; the job stays open



Personal details locked away

Names, phones and addresses in an encrypted vault; the house is just a hexagon

What we tested, and what we can't claim yet

Tested

- 80 automated checks on every code change: data, the reader, the planner, the server
- Five random years of requests; the result held in all five
- Every community's position checked against satellite photos
- Accessibility: 0 WCAG 2.2 AA issues in every view
- Security: locked-down site, validated inputs, sign-in rules ready

Not yet

- The repair requests are synthetic: no public NT repair data exists
- The reader is tested in English only. Tenants speak Kriol, Aboriginal English and many languages
- No community has reviewed it yet. That comes first, not last
- Our test kit is ready the day real reports arrive, labelled with the Aboriginal Interpreter Service

What we recommend



Make the trade-off a signed decision

Publish the triage rules, and record who decides how much cost may count, so remote waits are never an accident.



Pilot in one region, with communities

Start with the Katherine hub's 16 communities, co-designed with tenants, Aboriginal Housing NT, land councils and the Interpreter Service.



Test the AI on real words first

Score the reader on real reports in Kriol and Aboriginal English before it reads a single one on its own.

Where someone lives shouldn't decide how long they wait

ReachNT fixes 9 in 10 urgent remote repairs within 3 days instead of 62, for 44% more, and makes someone own that choice.

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Thank you. Questions welcome.

